Fiscal Year:	FY 2016	**Task Last Updated:**	FY 07/28/2016
PI Name:	Hammond, Timothy M.D.		
Project Title:	Yeast Colony Survival in Microgravity Depends on Ammonia Mediated Metabolic Adaptation and Cell Differentiation -Flight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	None		
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Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	08/01/2013	**End Date:**	07/31/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	Rapid Turn Around Flight ISS NOTE: Extended to 7/31/2018 per F. Hernandez/ARC (Ed., 7/31/17) NOTE: End date changed to 7/31/2017 per NSSC information (Ed., 7/25/16) NOTE: End date is 7/31/2016 (originally 7/31/2015) per NSSC information (Ed., 3/27/14)		
Key Personnel Changes/Previous PI:	May 2015 report: Added Drs. Corey Nislow and Guri Giaever as Co-Is.		
COI Name (Institution):	Birdsall, Holly M.D., Ph.D. (Baylor College of Medicine) Nislow, Corey Ph.D. (University of British Columbia) Giaever, Guri Ph.D. (University of British Columbia)		
Grant/Contract No.:	NNX13AN32G		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	Yeast chemical genomics is a powerful tool that is being applied to drug discovery. This approach allows the study of compound-target relationships in the context of a living cell with a systems biology view that includes both the cell target and the interdependent networks that respond to chemical stress. The modulation of yeast colonies, especially the redox potential, allows study of different drug pathways. This application investigates microbial metabolic adaptation and cell differentiation to the multiple stimuli encountered in space flight environments, using the yeast Saccharomyces cerevisiae as a model system. We have shown that microgravity activates a unique set of genes in yeast. Our previous work has largely defined yeast cellular stress defense response to microgravity, but stress defense responses are only involved in the survival of yeast in liquid cultures. It has recently been found that survival of plate-based yeast colonies is dependent on signaling mediated through ammonia which drives the multicellular yeast colony to differentiate into two populations, L or lower cells and U or upper cells. These two populations have different metabolic pathways and establish a cycle of nutrients that allows the U cells to continue to proliferate at the leading edge. Without ammonia signaling, the entire colony dies. Lack of convection in space is likely to dramatically impact signaling by ammonia, a volatile gas. We postulate that the flow of nutrient and ammonia signals will differ in microgravity where there is little to no convection. We hypothesize that yeast colony survival in microgravity depends on ammonia mediated metabolic adaptation and cell differentiation rather than on stress defense. To test our hypothesis, we will identify the mechanisms mediating colony stratification into U and L cells under multiple gravitational environments: microgravity emulated by Rotating Wall Vessel and Random Positioning Device, static one g cultures, and ultimately microgravity of space flight. Our experimental model will compare liquid and solid phase yeast cultures and make use of available mutant strains of yeast with deletion of the enzymes involved in ammonia signaling and the connected alternative metabolism activation, enzymes involved in stress responses, and the Msn4 promoter and SSA4 gene that we have previously identified as mediating response to microgravity. Relevant metabolic pathways will be studied using green fluorescent protein-linked reporter genes that can be analyzed by confocal microscopy of intact colonies, and flow cytometry of cells divided into L and U populations by sucrose density centrifugation. There are similarities between multicellular yeast colonies and tumors of clusters of mammalian cells in tissue culture. Thus insights into the effect of microgravity on cell to cell signaling may have far-reaching applications. Microgravity could provide a physiologically responsive test bed for drug pathway analysis at diverse redox states.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Yeast chemical genomics is a powerful tool that is being applied to drug discovery. This approach allows the study of compound target relationships in the context of a living cell with a systems biology view that includes both the cell target and the interdependent networks that respond to chemical stress. It has recently been found that survival of plate-based yeast colonies is dependent on signaling mediated through ammonia which drives the multicellular yeast colony to differentiate into two populations, L or lower cells and U or upper cells. Lack of convection in space is likely to dramatically impact signaling by ammonia, a volatile gas. There are similarities between multicellular yeast colonies and tumors of clusters of mammalian cells in tissue culture. Thus insights into the effect of microgravity on cell to cell signaling may have far-reaching applications. Microgravity could provide a physiologically responsive test bed for drug pathway analysis at diverse redox states.
Task Progress:	Task progress has been in three parts. Task progress part I. The effects of spaceflight on yeast have high concordance with agents that induce a very low intracellular redox state, and induce a massive efflux of glutathione. These results raise important issues. Can the reduced redox state during spaceflight be reproduced and modulated in ground-based simulations? Will this allow definition of unique drug pathways, as a low redox state mirrors the electrophilic properties of mitochondria where many drugs are metabolized? Unfortunately, assays for redox status and its major cellular determinant, glutathione, are diverse and often cell type specific. Currently, an accepted redox probe set for yeast studies is not available. This paper validates fluorescent probes for glutathione and reactive oxygen status in yeast to support mechanistic studies of microgravity and drug metabolism. The plethora of fluorescent reagents for reactive oxygen species and glutathione makes head-to-head comparisons of all the alternatives impractical. These reagents measure the physiological milieu of reactive oxygen species and diverse thiols, rather than specific individual molecules. We report that in yeast, monochlorobimane (mBCL) and 2',7'-dichlorodihydrofluorescein diacetate (DC-FDA) are suitable for fluorometric and flow cytometry studies of glutathione and reactive oxygen species, respectively. Both dyes have low background fluorescence, predictable loading, good retention, and are non-toxic to Saccharomyces cerevisiae. Both dyes show concordance with other fluorescent and biochemical assays of reactive oxygen species. Task progress part II. Comparison of two powerful investigative models for drug pathways using Saccharomyces cerevisiae are compounded with the use of different growth media and stimuli to the system. Chemogenetic profiling uses a pool of yeast deletion mutants to determine survival changes. These studies are heavily biased to use of rich media, and have been used to study >3,000 small molecules including many drugs. Giant yeast colonies studies are heavily biased to use of poor media. Ammonia secretion is the predominant stimulus to get giant yeast colonies to differentiate. The current study answers two questions. First, what is the difference in metabolic status over time between giant yeast colonies grown on rich and poor media? Metabolic status is assayed as glutathione content, redox potential state, cell death, ATP content, and NAD:NADH and NADP:NADPH ratios. Second, how does media choice affect physiological responses of giant yeast colonies? We chose rotation as the physiological stimulus, as rotation randomizes the gravitational vector, which drives convection medicated ammonia signaling. This study demonstrates that redox potential status can be changed over time in giant yeast colonies using rich media. Second, ammonia disbursement by rotation of giant yeast colonies provides a physiologically intact method to manipulate redox potential status in giant yeast colonies. Task progress part III. This application investigates microbial metabolic adaptation and cell differentiation to the multiple stimuli encountered in spaceflight environments, using the yeast Saccharomyces cerevisiae as a model system. We have shown that microgravity activates a unique set of genes in yeast. Our previous work has largely defined yeast's cellular stress defense response to microgravity, but stress defense responses are only involved in the survival of yeast in

	liquid cultures. It has recently been found that survival of plate-based yeast colonies is dependent on signaling mediated through ammonia which drives the multicellular yeast colony to differentiate into two populations, L or lower cells and U or upper cells. These two populations have different metabolic pathways and establish a cycle of nutrients that allows the U cells to continue to proliferate at the leading edge. Without ammonia signaling, the entire colony dies. Lack of convection in space is likely to dramatically impact signaling by ammonia, a volatile gas. We postulate that the flow of nutrient and ammonia signals will differ in microgravity where there is little to no convection. We hypothesize that yeast colony survival in microgravity depends on ammonia mediated metabolic adaptation and cell differentiation rather than on stress defense. To test our hypothesis, we identified the mechanisms mediating colony stratification into U and L cells under multiple gravitational environments: microgravity emulated by Rotating Wall Vessel and Random Positioning Device, static one g cultures, and ultimately microgravity of space flight. Our experimental model compared liquid and solid phase yeast cultures and made use of available mutant strains of yeast with deletion of the enzymes involved in ammonia signaling and the connected alternative metabolism activation, enzymes involved in stress responses, and the Msn4 promoter and SSA4 gene that we have previously identified as mediating response to microgravity. Relevant metabolic pathways were studied using green fluorescent protein-linked reporter genes that can be analyzed by confocal microscopy of intact colonies, and flow cytometry of cells divided into L and U populations by sucrose density centrifugation. There are similarities between multicellular yeast colonies and tumors of clusters of mammalian cells in tissue culture. Thus insights into the effect of microgravity on cell to cell signaling may have far-reaching applications, as microgravity could provide a physiologically responsive test bed for drug pathway analysis at diverse redox states We proposed to simply progress from targeted gene and promoter study to genome wide studies. Specifically there were 48 samples for imaging as previously planned, progress from target gene analysis to genome wide study including three yeast deletion series: one wild type, and two double deletion on SOK-2 (ammonia), and TRR1 (redox) deletion backgrounds. Analysis by next generation sequencing is progressing to apply all the power of modern genomics analysis. All samples were imaged on an InCell fluorescent imager, and fluorescent strains quantitatively analyzed in ~600 flow cytometry runs. This approach has the advantages that it fulfills the origin aims of grant, allows genome wide analysis for sophisticated pathway analysis, has phenotypic outcome measures by using survival advantage in the yeast deletion pool as a met. This approach using genome wide analysis contributes to the NASA OMICS initiative, directly addresses Decadal Survey Report Recommendation P2, and directly address the aims of the NASA GeneLab initiative.

Bibliography Type:	Description: (Last Updated: 03/31/2026)
Articles in Peer-reviewed Journals	Hammond TG, Allen PL, Birdsall HH. "Validation of assays for reactive oxygen species and glutathione in Saccharomyces cerevisiae during microgravity simulation." Gravitational and Space Research. 2015 Dec;3(2):42-53. http://gravitationalandspacebiology.org/index.php/journal/article/view/701/732 , Dec-2015
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Articles in Peer-reviewed Journals	Birdsall HH, Allen PL, Hammond JS, Gunter MA, Hammond TG. "Establishing a low redox potential in giant yeast colonies: effects of media and rotation." Gravitational and Space Biology. 2016 Jul;4(1):27-38. http://gravitationalandspacebiology.org/index.php/journal/article/view/718 , Jul-2016